Home Reading 09 — Optional Root Locus and Robustness

Before Class

Preview Lecture 9

Optional: read Lecture 9: Root Locus and Robustness.

Textbook Reading

- Ogata, System Dynamics, 4th edition, Section 10-8.

This Week Introduces

Closed-loop pole movement with gain, response trade-offs, actuator demand and robustness to plant uncertainty.

Tutorial Preparation

Read Tutorial 9 before class. Be ready to:

- Q1: Interpret root-locus information and check whether a command is feasible.
- Q2: Design a velocity-feedback inspection stage.
- Q3: Test a nominal design against uncertain damping.

Check Your Understanding Before Class

Before class, try to answer these two questions:

- Why do closed-loop poles move when controller gain changes?
- Why must a gain satisfy response, actuator and uncertainty limits at the same time?

Prof. Zengxi (Stephen) Pan | University of Wollongong | TYUT 2026
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